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Knowledge - based algorithm

Function TT_ENTAILES? (KB, α) return true or false
inputs: KB, the knowledge base, a sentence in propositional logic
α, the query, a sentence in propositional array
Symbols ← a list of the proposition symbols in KB and α
return TT - check - All (KB, α, symbols, {})
Function TT- check_all (KB, α, symbols, model) return true or: false

if empty? (symbols) then
if PL-true? (KB, model) then return PL-true? (α, model)

else return true
else do
P ← first
rest ← rest
return (TT_check_all (KB, α, rest, model ( ) ∪ {P=true})
and TT_check_all (KB, α, rest, model ( ) ∪ {P=false}))

i) Truth table

P	Q	R	Q→P	P→¬Q	Q∨R	KB
F	F	F	T	T	F	F
F	F	T	T	T	T	T
F	T	F	F	T	T	F
F	T	T	F	T	T	F
T	F	F	T	T	F	F
T	F	T	T	T	T	T
T	T	F	T	F	T	F
T	T	T	T	F	T	F

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models of KB:

(P=F, Q=F, R=T)
(P=T, Q=F, R=T)

(ii) Yes

In every model where KB is true.
R=T

KB ⊨ R

(iii) NO

P=F, Q=F, R=T

KB is true but R → P is false

KB ⊭ R → P

(iv) Yes

where KB is true, Q=F

Therefore Q → R is true

KB ⊨ Q → R